Introduction to Monte-Carlo and cluster-expansion based simulations

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Aarhus DFT-ML workshop

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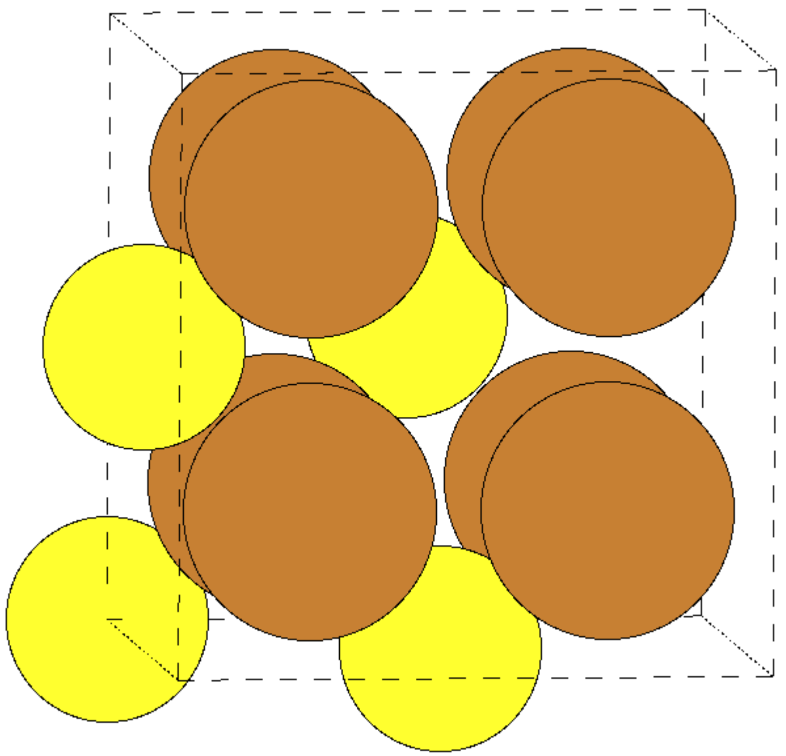
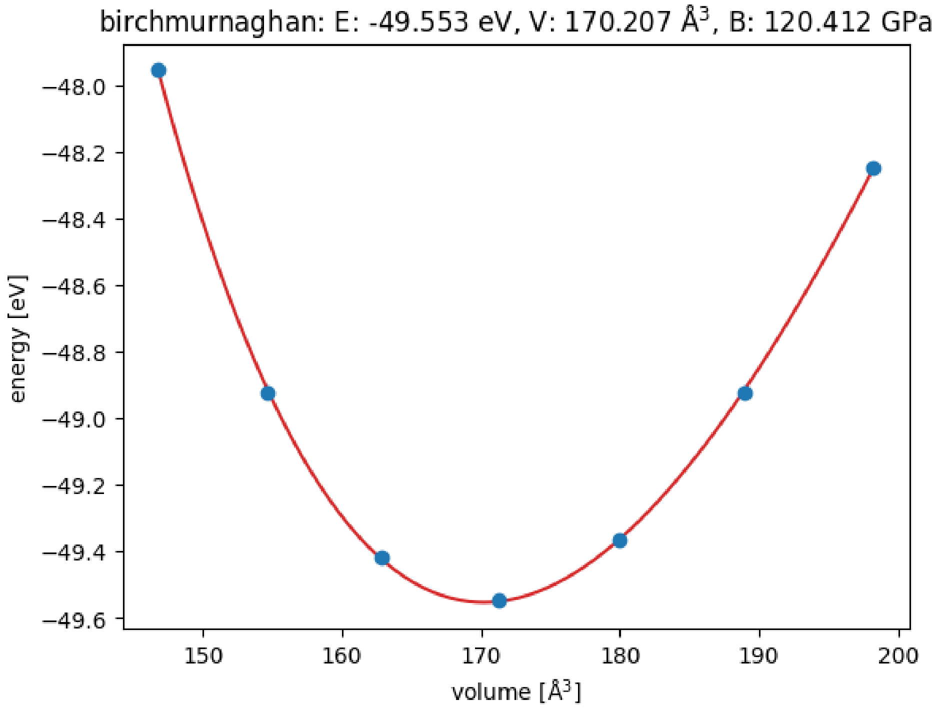
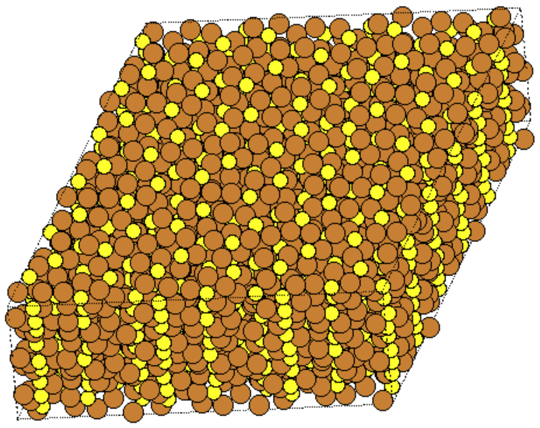
Recap: energy minimization

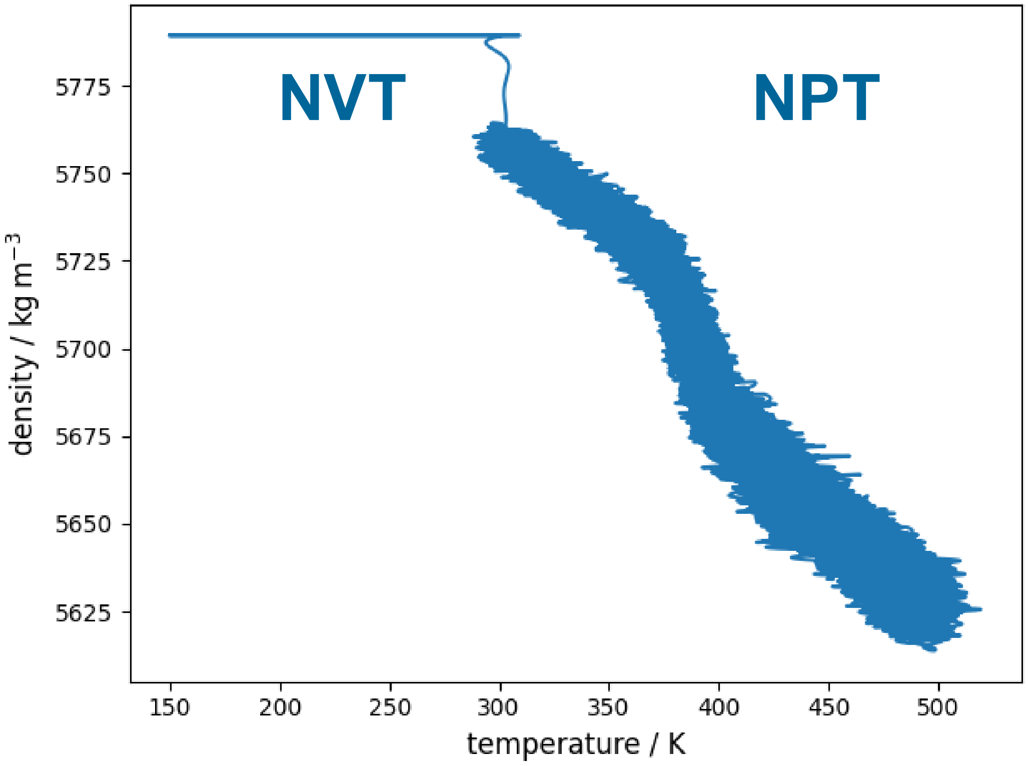


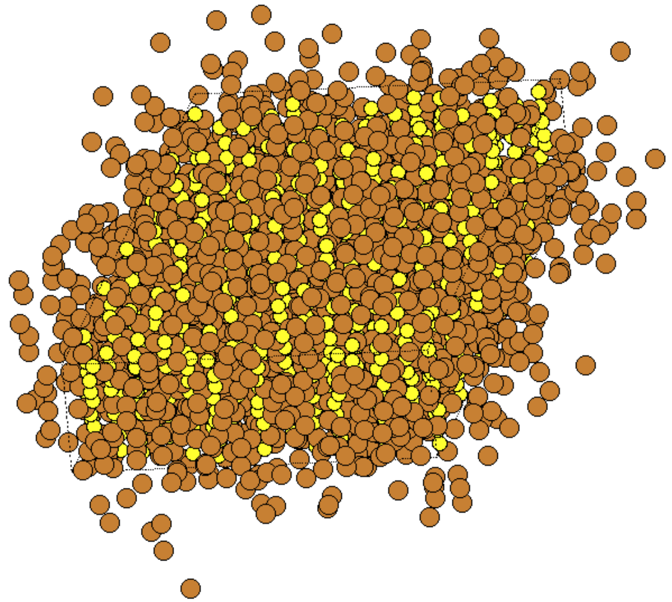


simulation at 0 K

Recap: Molecular dynamics



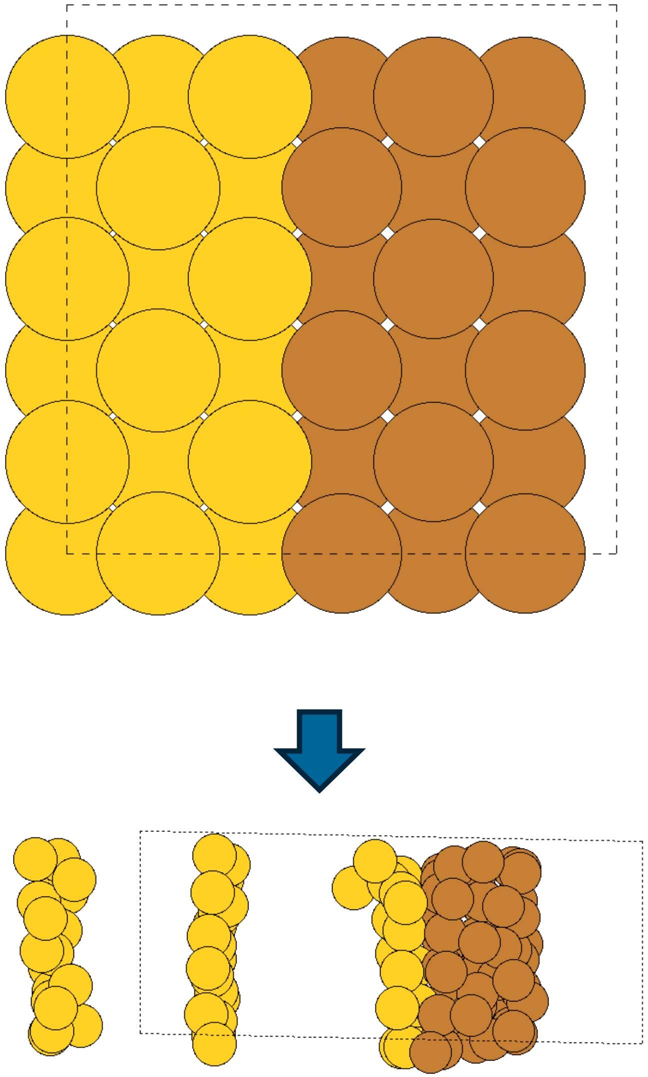




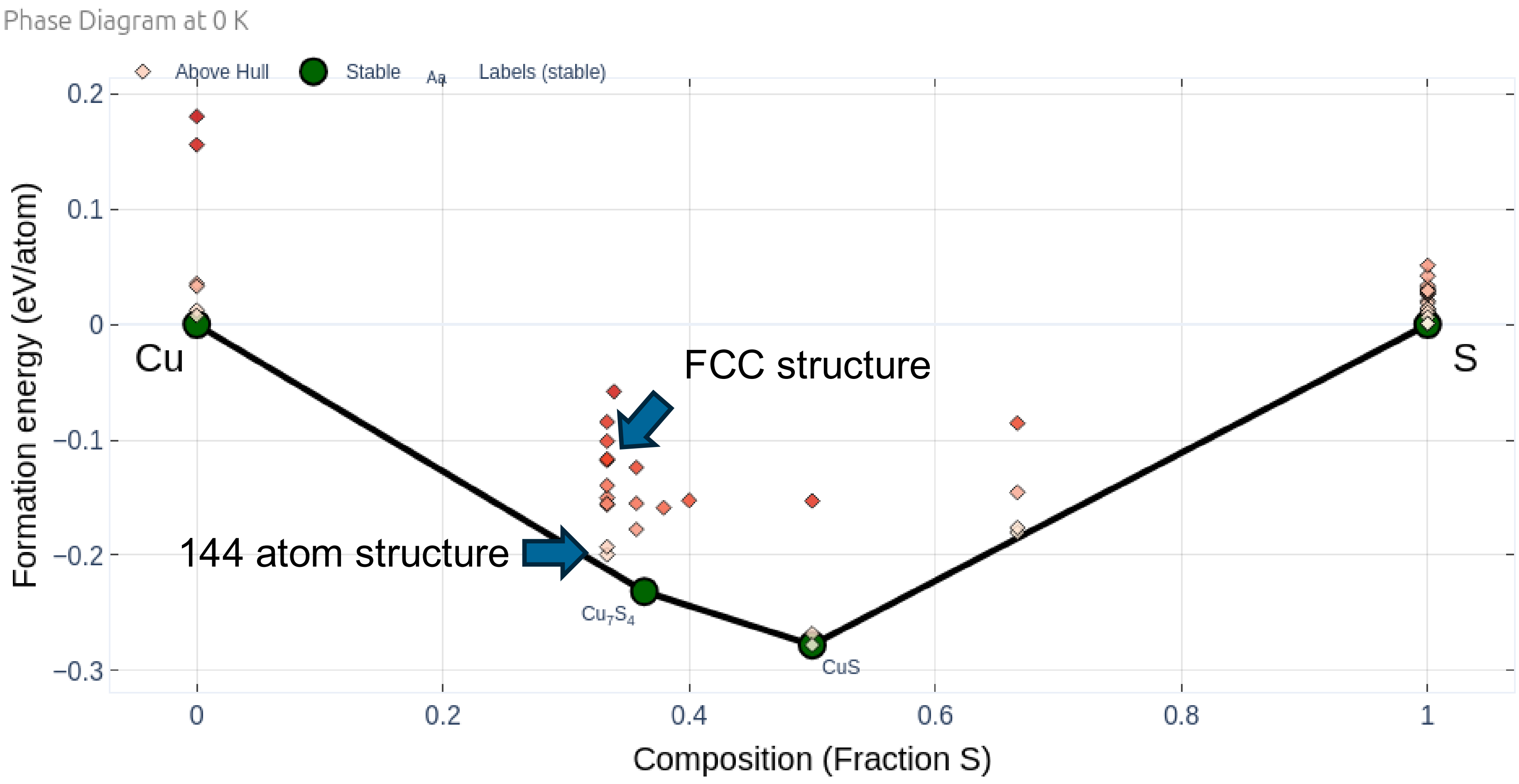
Difficulties with sampling

- Usually, the N (NVE, NVT, NPT) is constant – removing and adding particles is challenging in MD.
- Additionally, the starting configuration is crucial – phase transitions can be very difficult to observe.
- However, also concentration is usually constant, what if we are interested in stable compounds?

- Possibility: very large simulation cells, phases separate (**expensive**)



About thermal stability – starting at 0 K

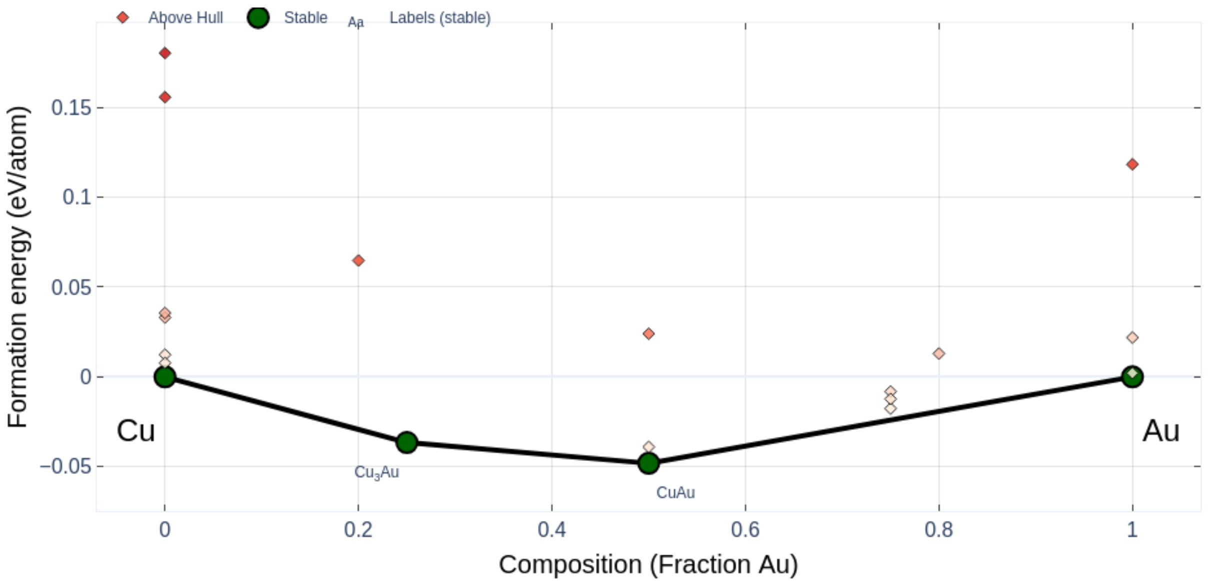


Source: https://next-gen.materialsproject.org/materials/mp-12087

Convex Hull

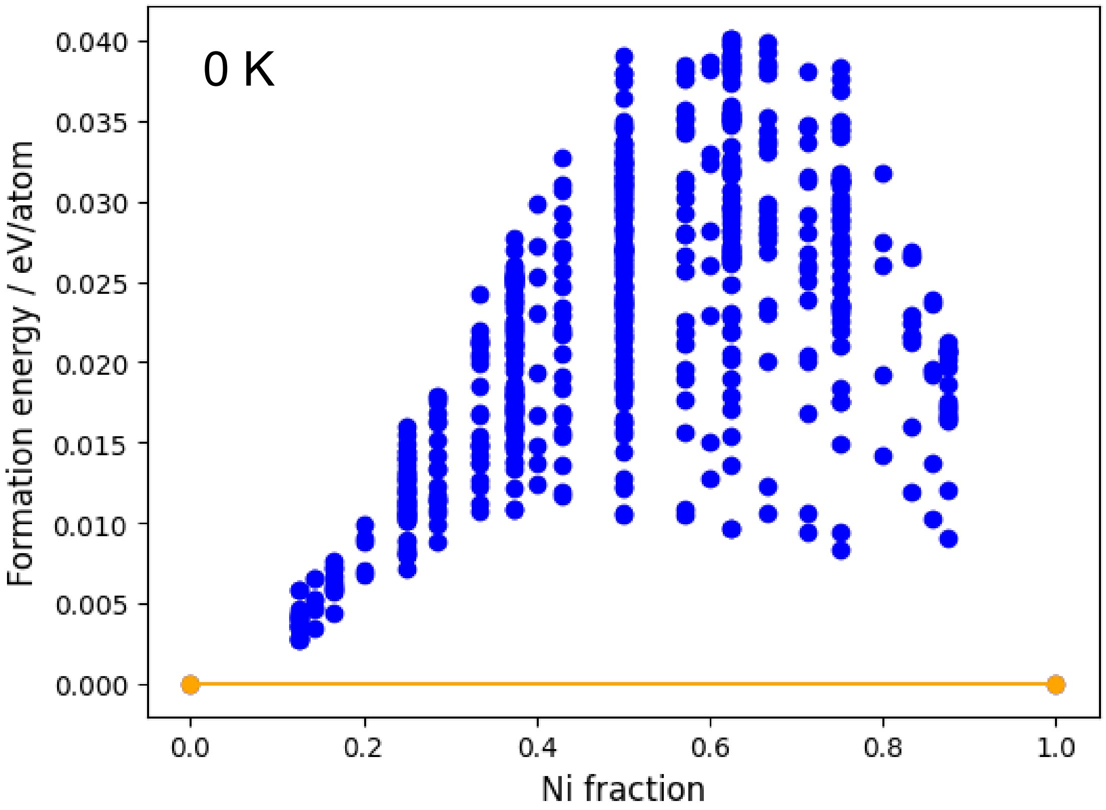
Structure above convex hull:
Decomposes into portions of other phases.

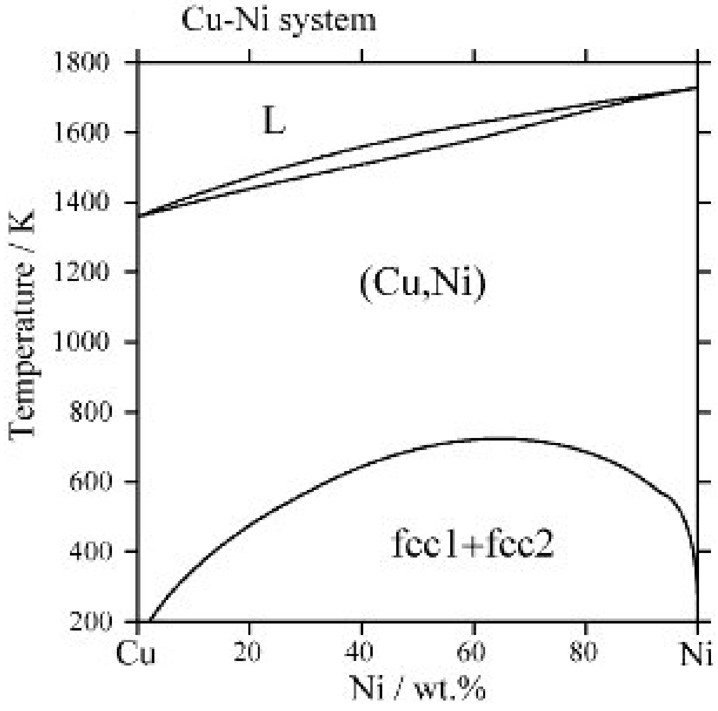
Materials Project can show these plots for most 2-element compounds



https://next-gen.materialsproject.org/materials/mp-522

More extreme example: CuNi



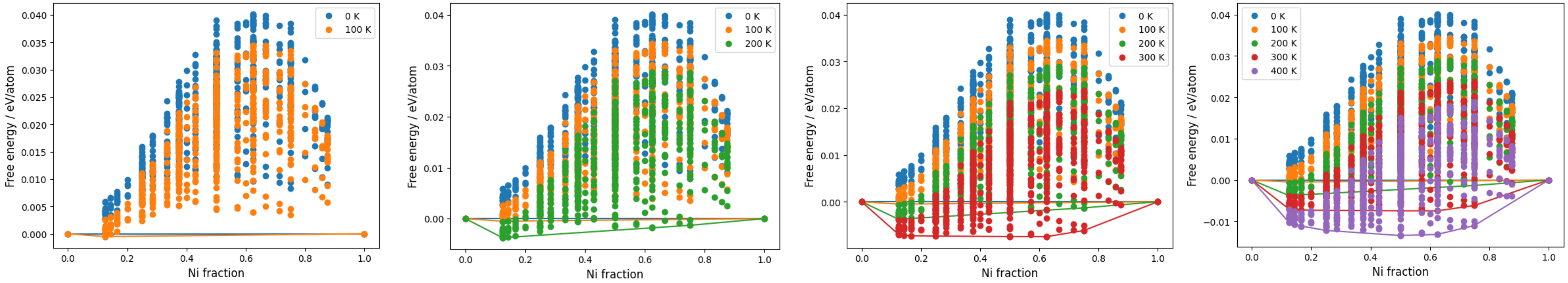


Taken from: ***Nanoscale***, 2014,**6**, 12490-12499

Adding temperature

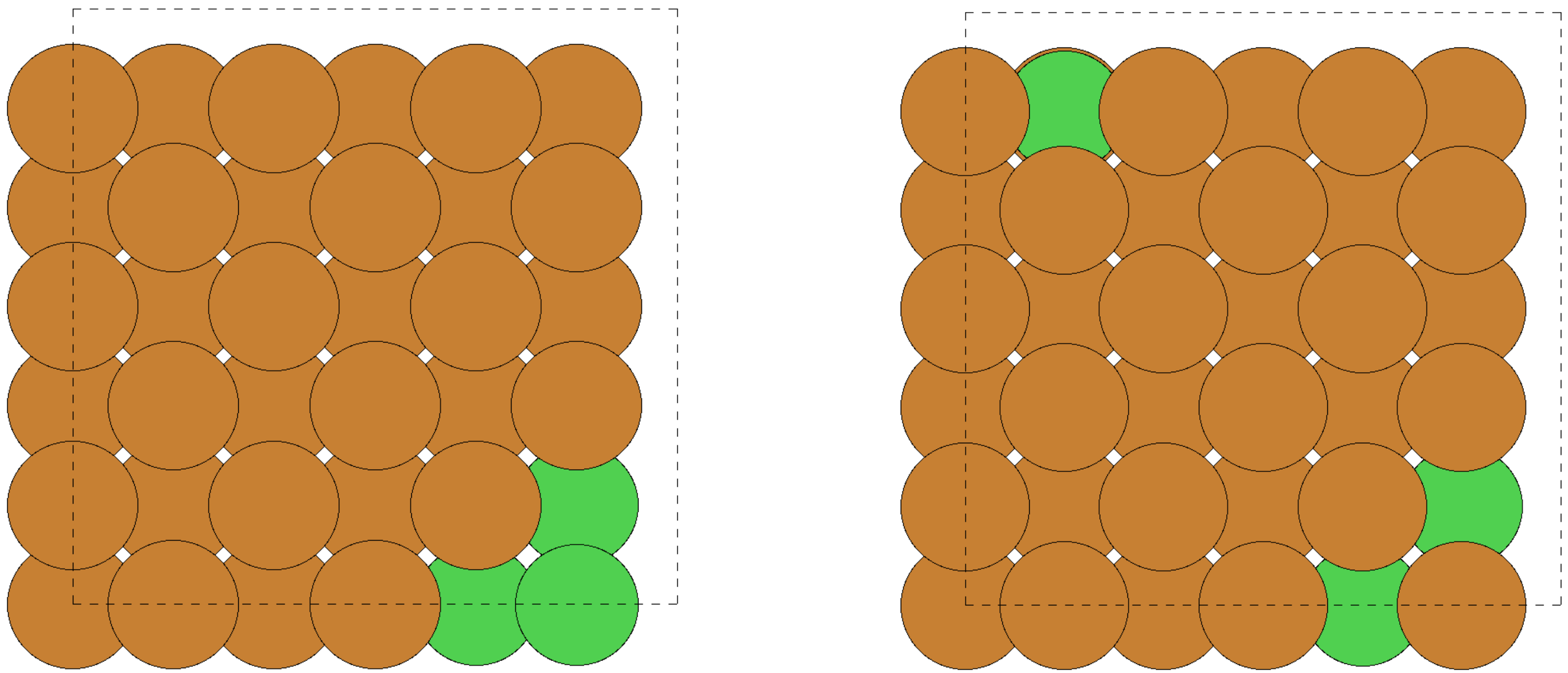
Miscibility gap becomes smaller at higher temperatures

Free energy: $F = E + pV - TS$



Ideal entropy of mixing: $\Delta S_{\text{mix}} = k_B(x_A \ln x_A + x_B \ln x_B)$

Monte Carlo simulations



Metropolis-Hastings

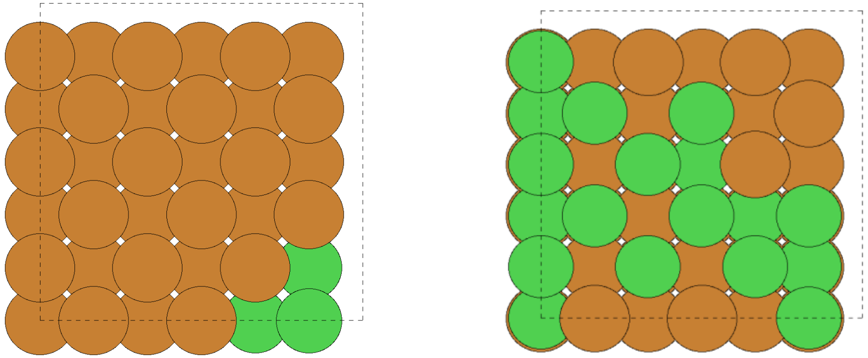
Acceptance probability: $\mathcal{P} = \min\{1, \exp(-\beta \Delta\psi)\}$

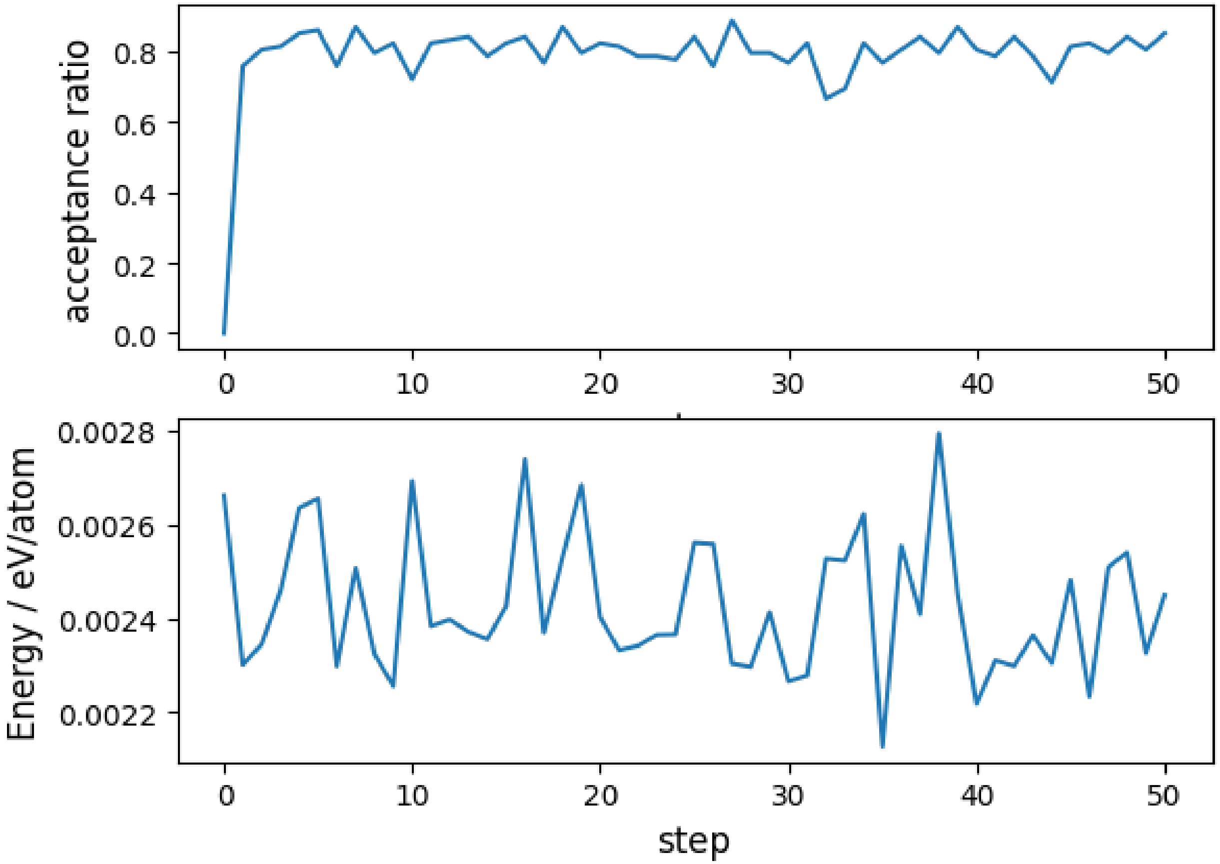
$$\beta = \frac{1}{k_B T}$$

Canonical ensemble

$$\mathcal{P} = \min\left\{1, \exp(-\beta\Delta\psi)\right\}$$

$$\psi = E$$

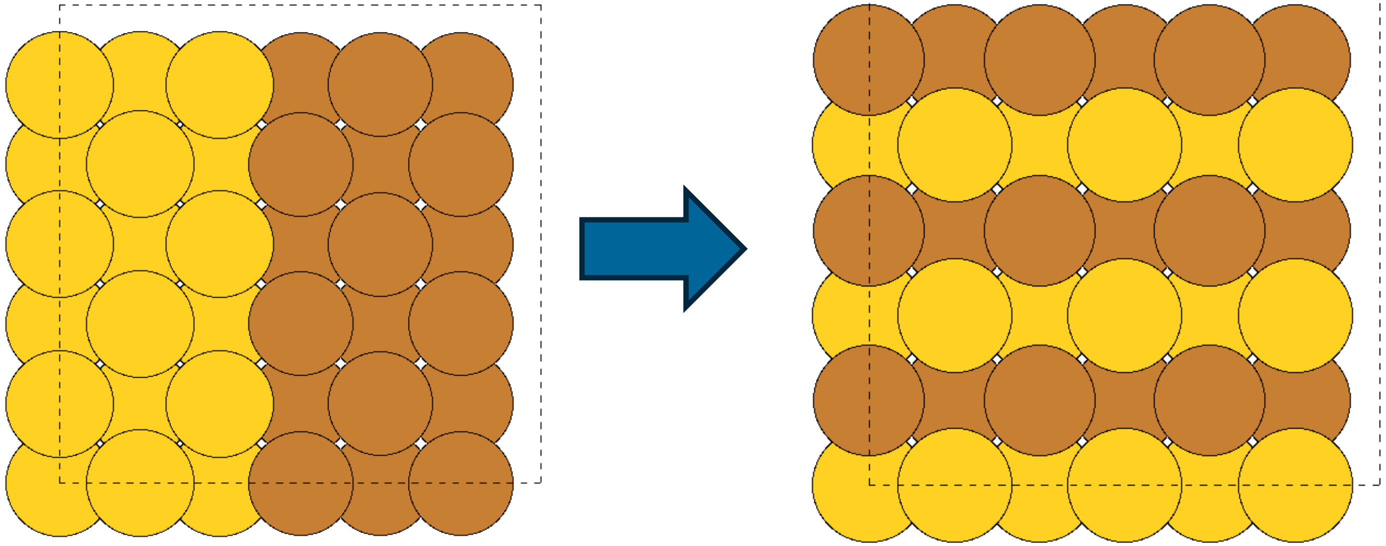


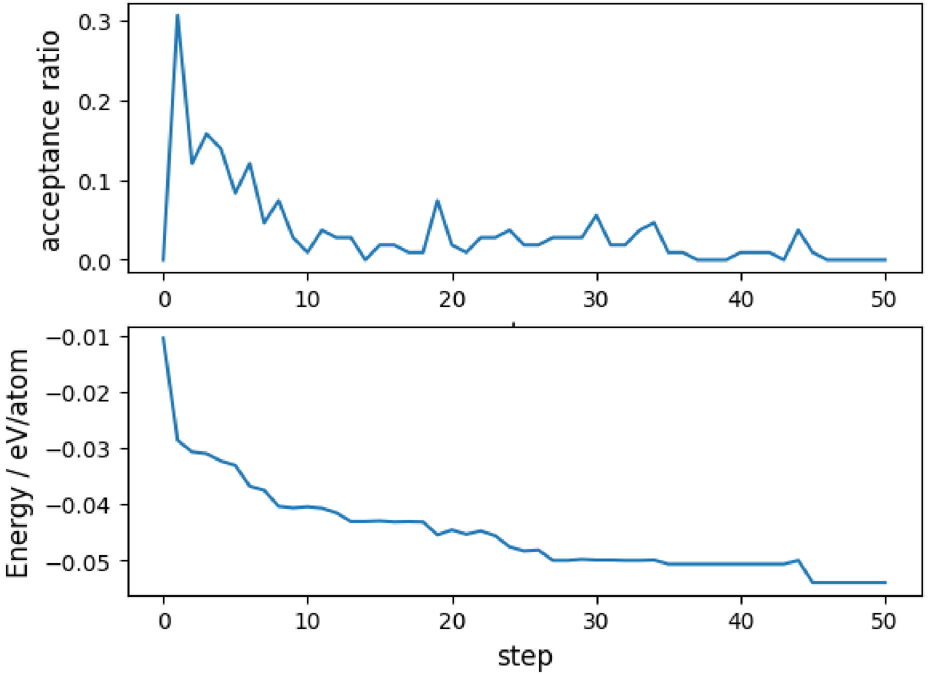


Allows to swap species, but the concentration is still constant

Canonical ensemble: CuAu

Concentration at 0.5





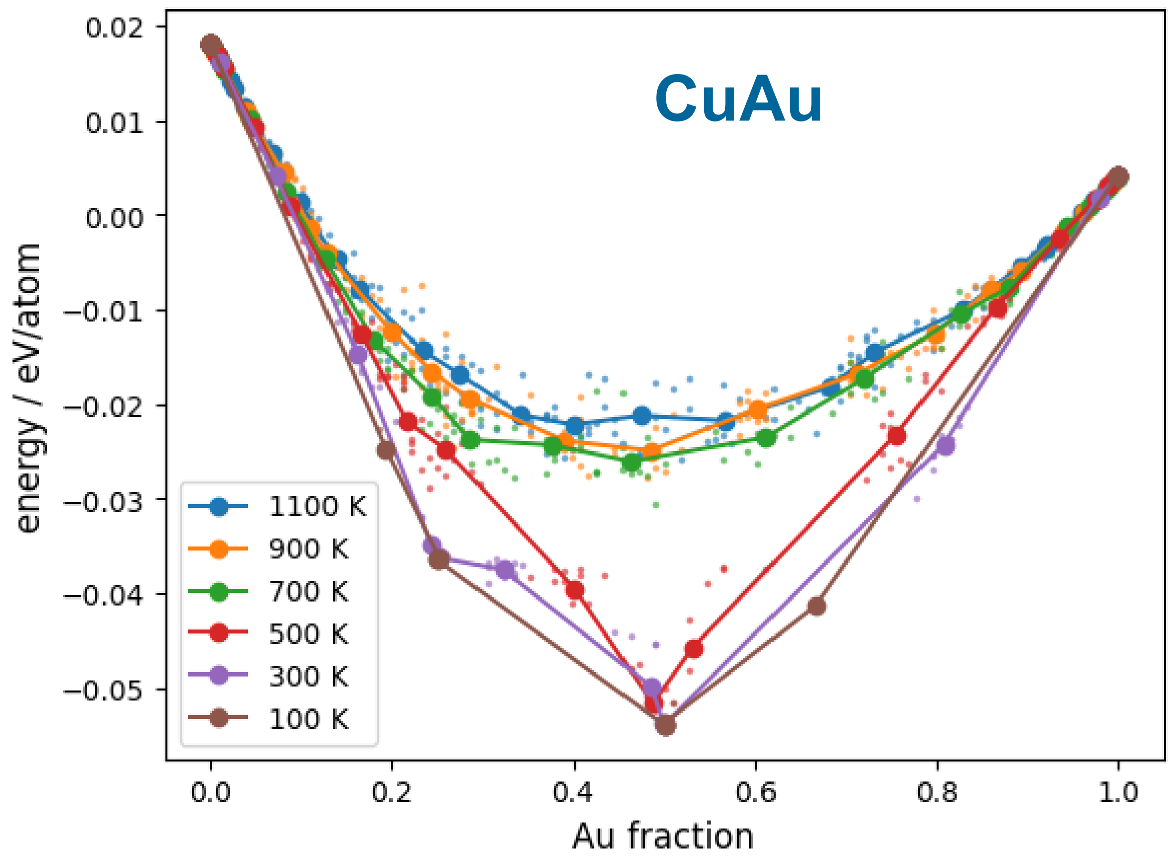
Semi-grand canonical ensemble

$$\mathcal{P} = \min\left\{1, \exp(-\beta\Delta\psi)\right\}$$

$$\psi = E - N\sum_{i=2} c_i \Delta\mu_i$$

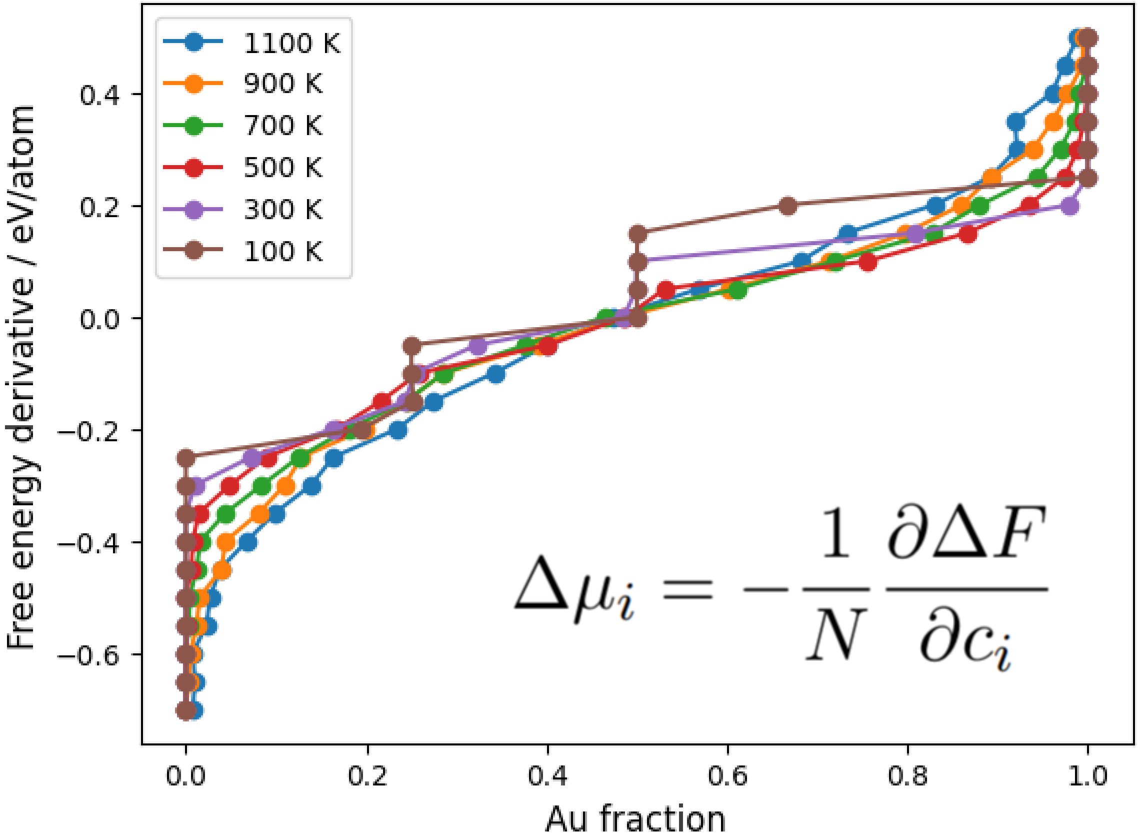
Chemical potential μ_i

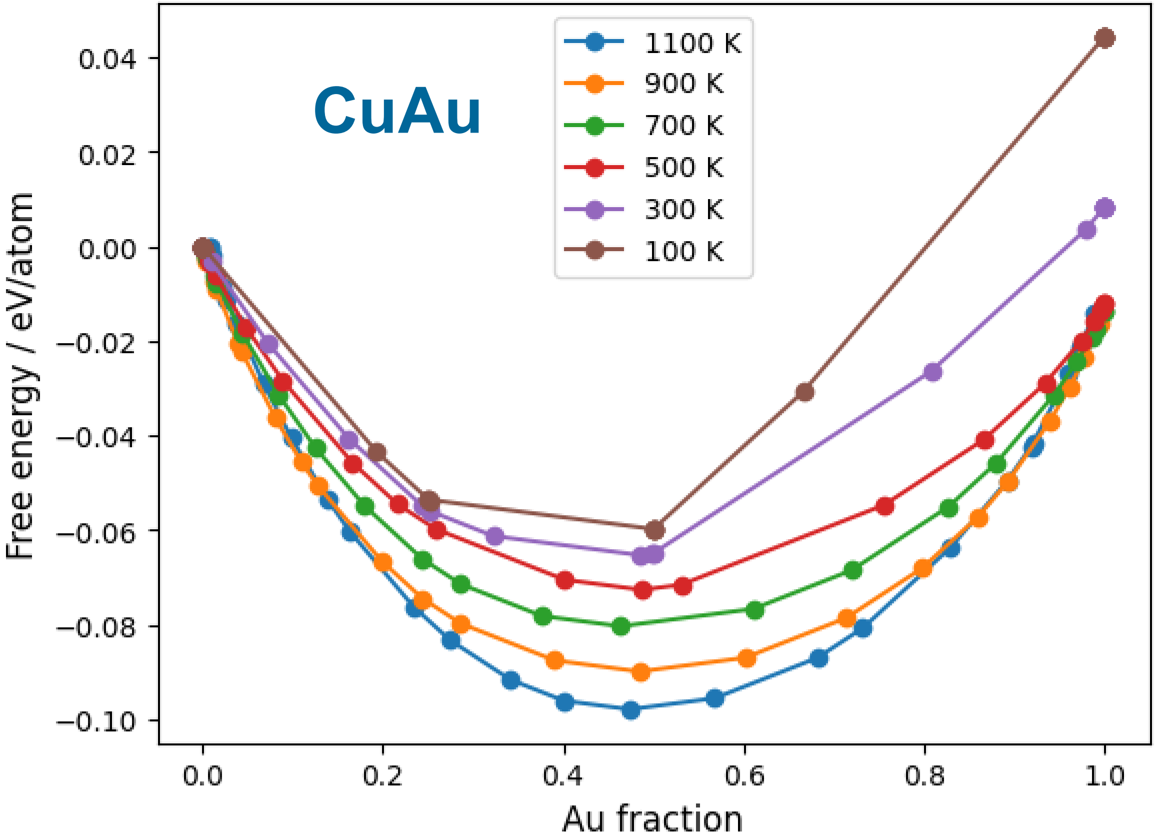
$$\Delta\mu_i = -\frac{1}{N}\frac{\partial \Delta F}{\partial c_i}$$



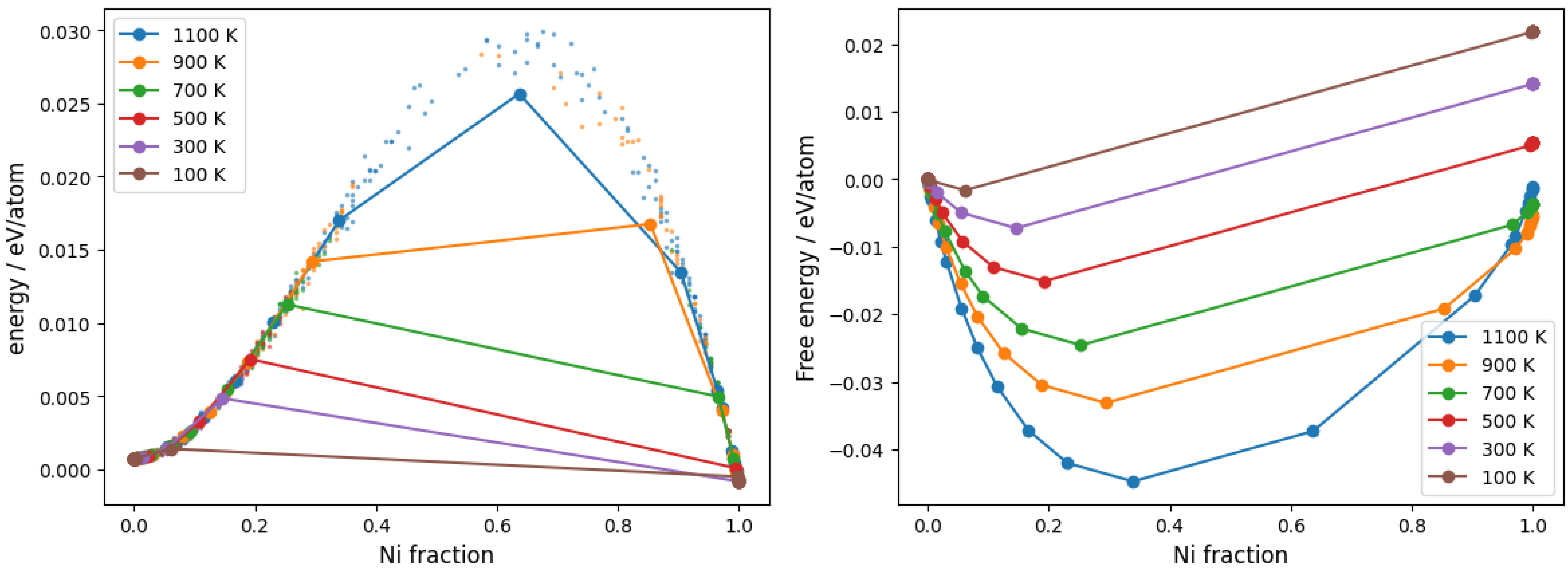
Allows dynamic concentrations

Free energy derivative





CuNi in SGC



SGC cannot sample over miscibility gaps

Variance-constrained semi-grand canonical ensemble

$$\mathcal{P} = \min\left\{1, \exp(-\beta\Delta\psi)\right\}$$

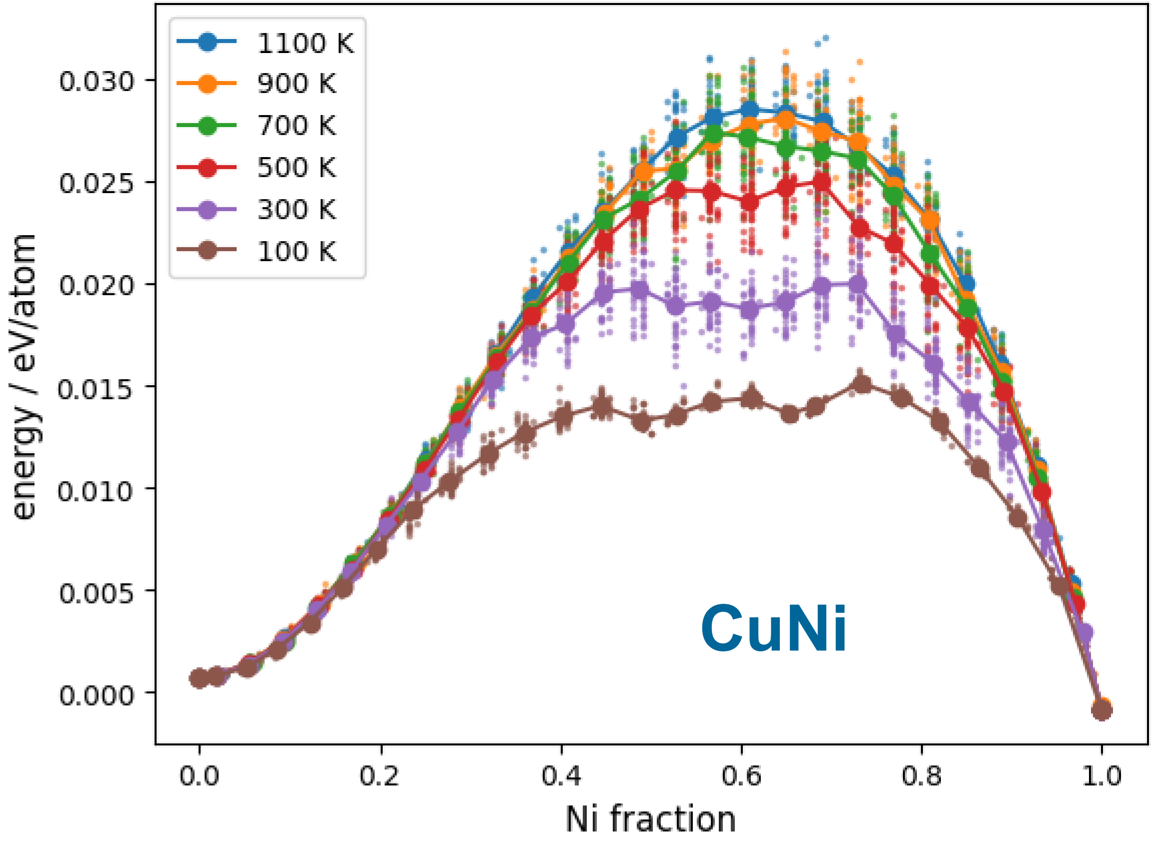
$$\psi = E + N k_B T \bar{\kappa}(c + \bar{\phi}/2)^2$$

$\bar{\phi}$ … constrains average of concentration

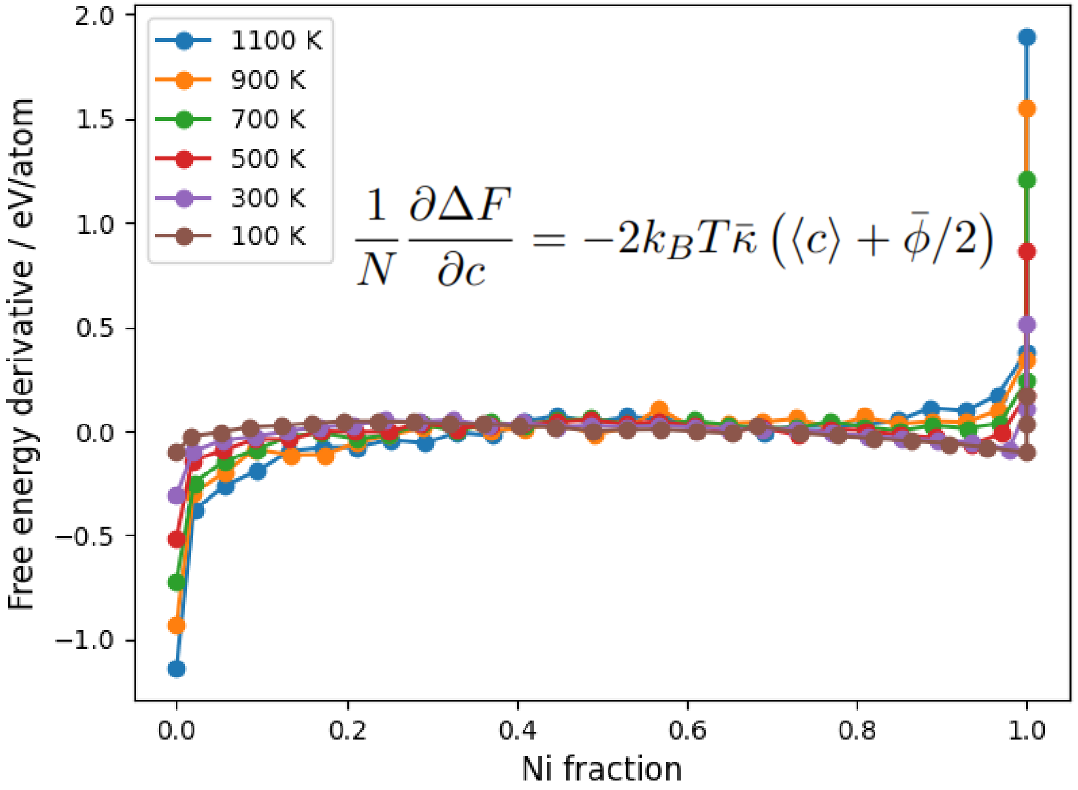
$\bar{\kappa}$ … constrains variance of concentration

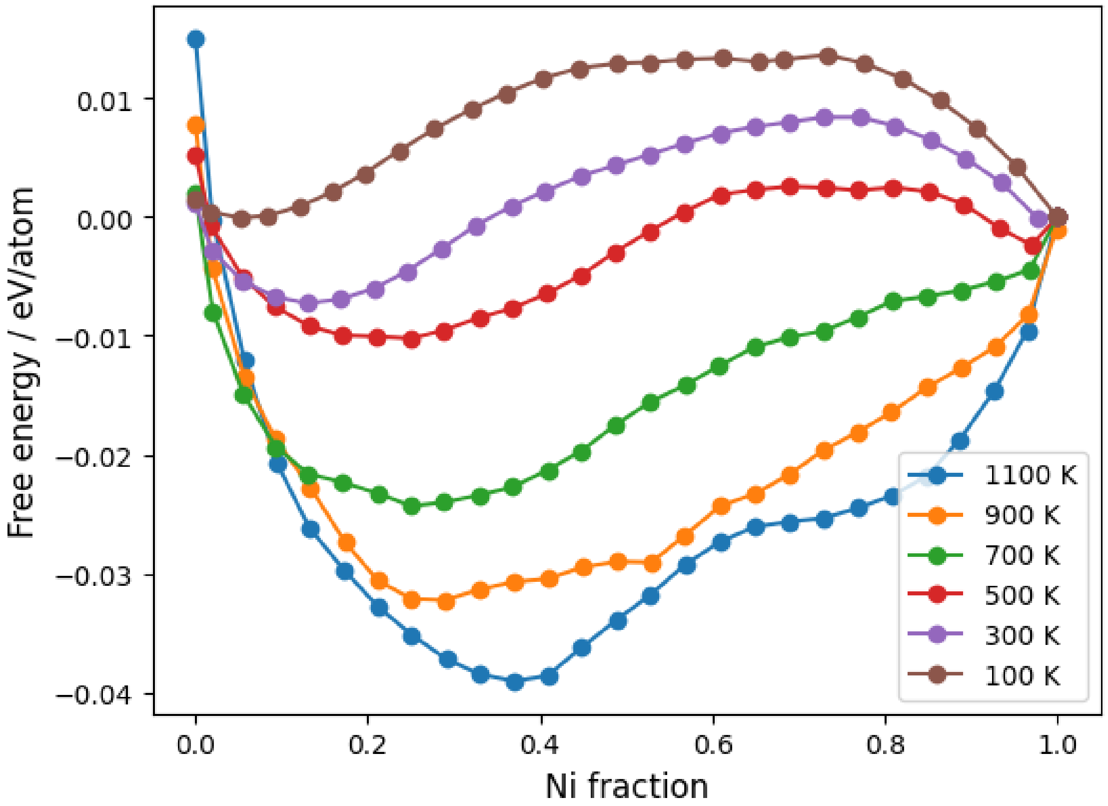
Allows sampling across miscibility gaps

Phase transition region can be seen in shape

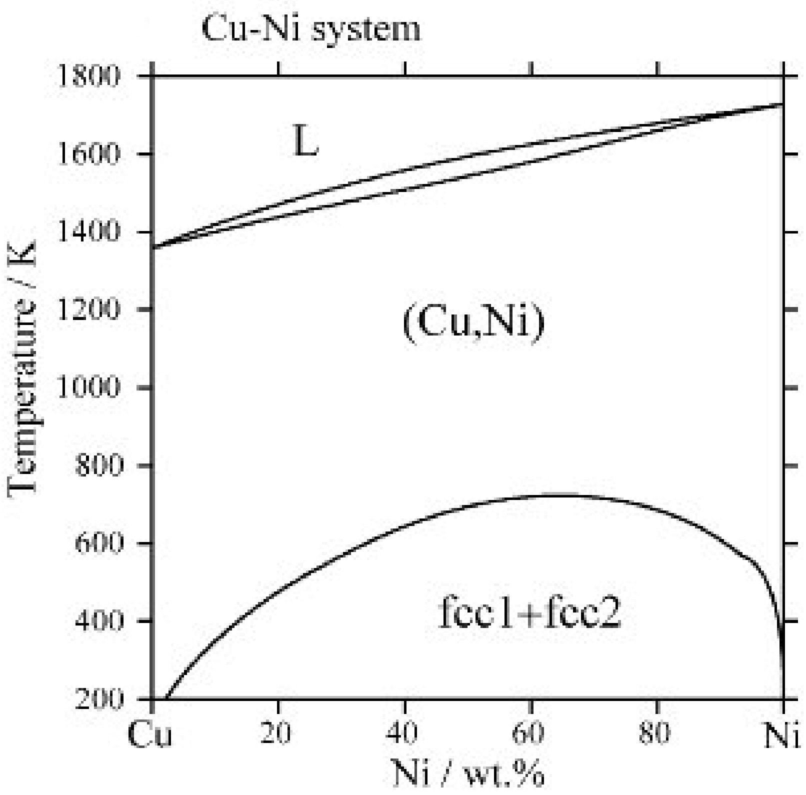


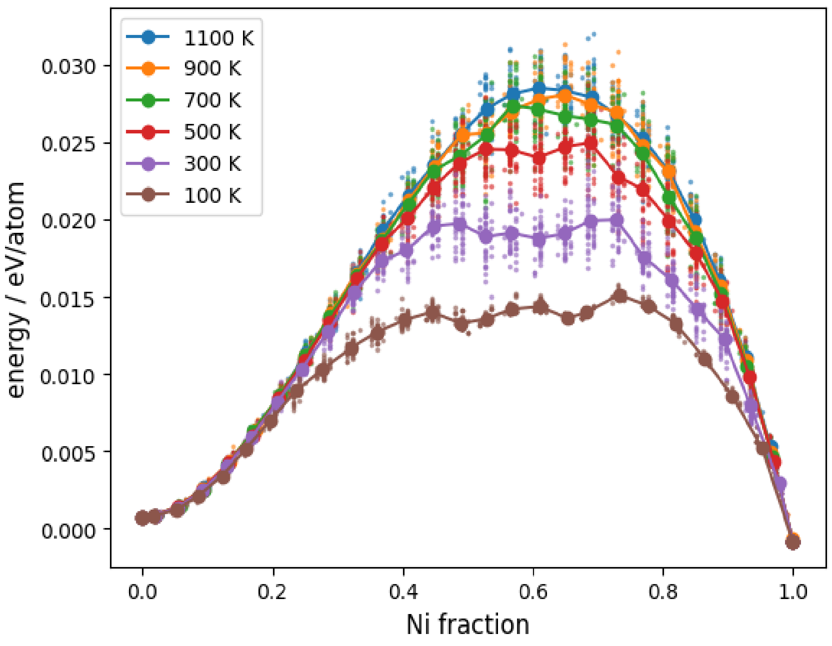
Free energy derivative (CuNi)

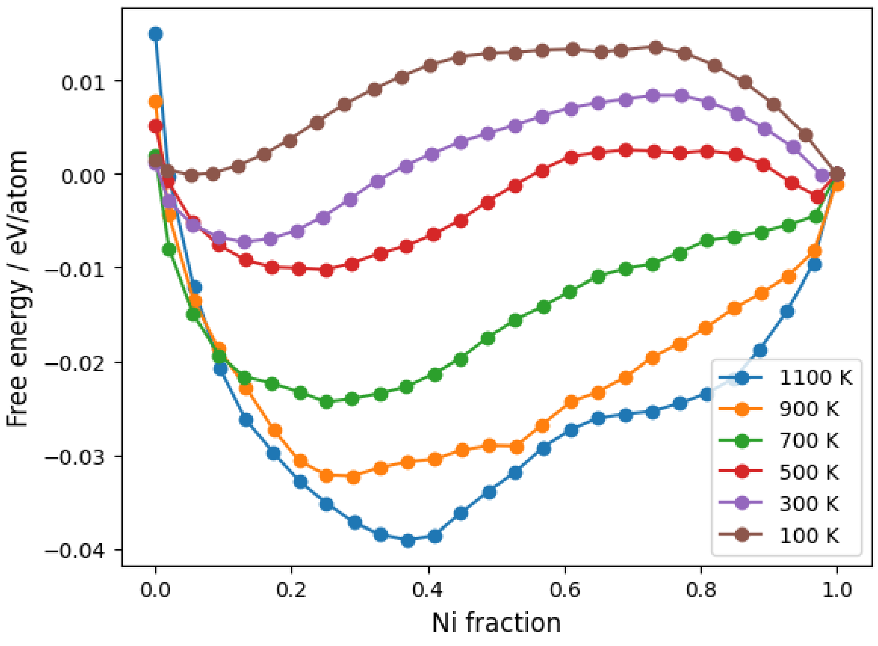




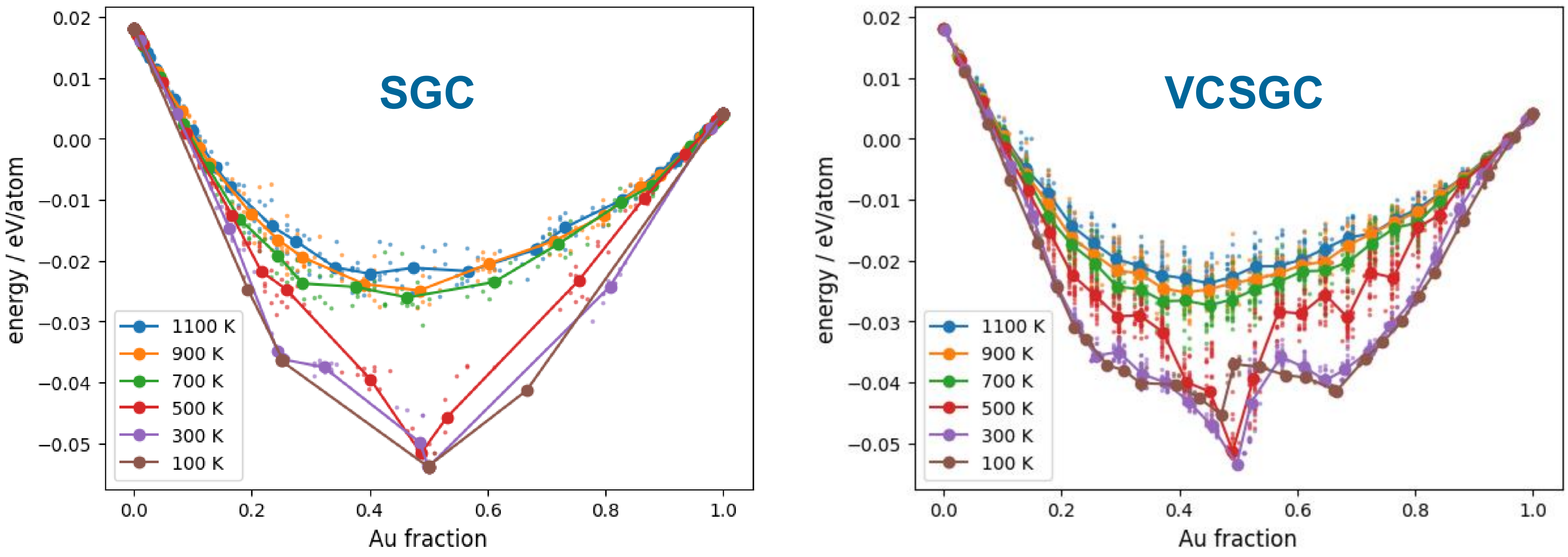
Comparison to phase diagram







VCSGC for CuAu



MC conclusion

- Allows sampling thermodynamics without passing the transitions
- Can provide insights about thermal stability of specific compositions
- Especially including the swapping can require a significant amount of steps for each of which the equilibrium structure and energy has to be determined
- With DFT and sometimes even with models such as MACE: **very expensive**

Methods to predict atomistic energies

QM methods

Beyond DFT

hybrid

meta GGA

DFT

GGA

LDA

Machine learning interatomic potentials

GAPs[5]

GRACE[3]

ACE[4]

MACE[2]

MTPs[8]

NEPs[1]

NequIP[7]

Classical force fields

LJ

Buckingham

Class 2/1 FFs[6]

speed

[1] 10.1103/PhysRevB.104.104309
[2] arXiv:2206.07697
[3] 10.1103/PhysRevX.14.021036
[4] 10.1103/PhysRevB.99.014104
[5] 10.1103/PhysRevMaterials.4.093802
[6] 10.1016/j.fluid.2006.07.014
[7] 10.1038/s41467-022-29939-5
[8] 10.1088/2632-2153/abc9fe

Ordering is a rough qualitative estimate and depends on many factors

Predicting optimized energies?

s

FT

GGA

LDA

Machine learning interatomic potentials

GAPs[5] GRACE[3]
ACE[4]
MACE[2] MTPs[8]
NEPs[1]
NequIP[7]

Classical force fields

LJ
Buckingham
Class 2/1 FFs[6]

Cluster Expansion

speed

[1] 10.1103/PhysRevB.104.104309
[2] arXiv:2206.07697
[3] 10.1103/PhysRevX.14.021036
[4] 10.1103/PhysRevB.99.014104
[5] 10.1103/PhysRevMaterials.4.093802
[6] 10.1016/j.fluid.2006.07.014
[7] 10.1038/s41467-022-29939-5
[8] 10.1088/2632-2153/abc9fe

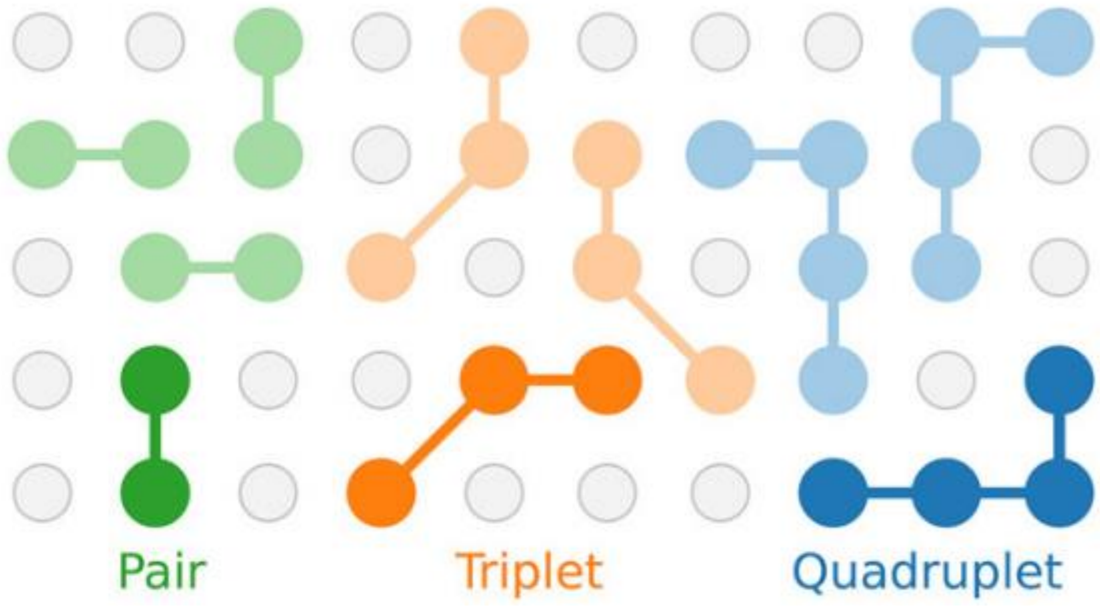
Ordering is a rough qualitative estimate and depends on many factors

Cluster expansion

Predicting a function $Q(\sigma)$
from a set of clusters

$$Q(\sigma) = Q_0 + \sum_{\alpha} \langle \Pi_{\alpha'}(\sigma) \rangle_{\alpha} \, m_{\alpha} J_{\alpha}$$

J_{α} … parameters (ECIs)
m_{α} … multiplicity
$\Pi_{\alpha'}(\sigma)$ … basis functions
Q_0 … configuration invariant term



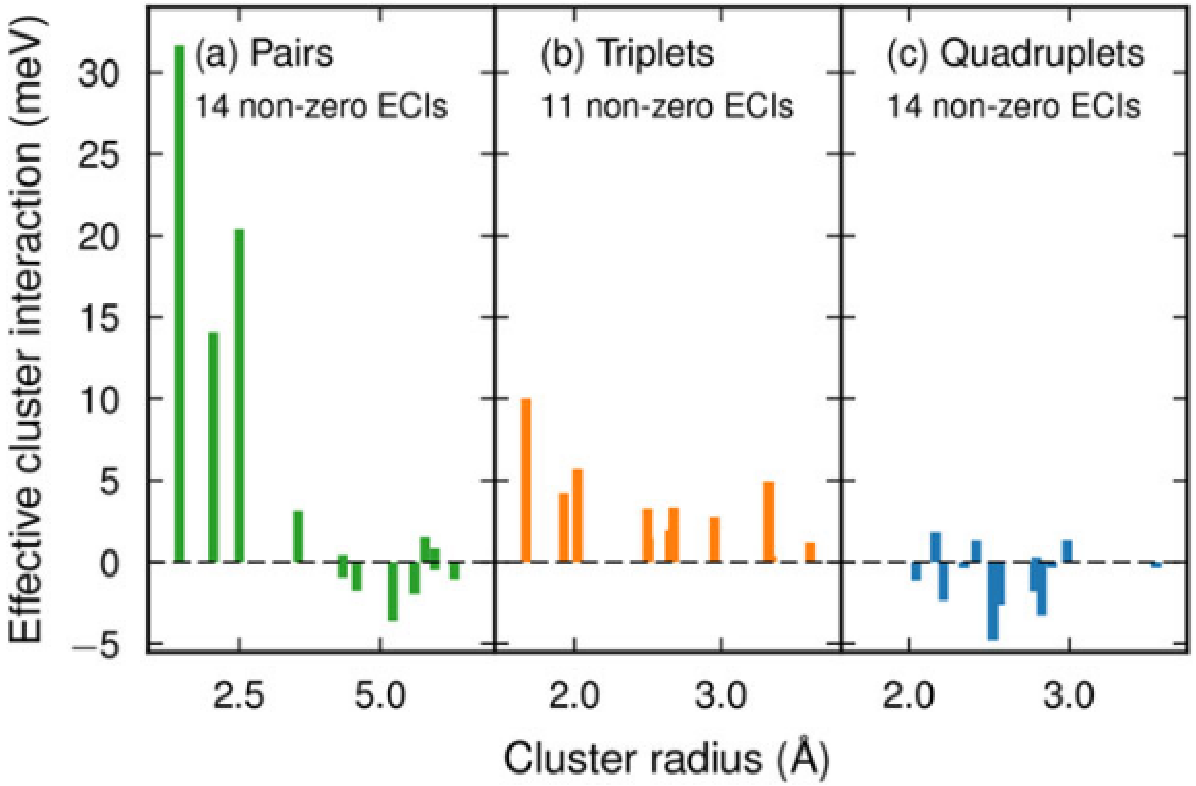
M. Ångqvist *et al.*, "ICET – A Python Library for Constructing and Sampling Alloy Cluster Expansions," *Advanced Theory and Simulations*, vol. 2, no. 7, p. 1900015, 2019, doi: 10.1002/adts.201900015.

Training the models

In our case, we would want to train on energies of mixing (or a formation energy)

CEs are much more simple than MLIPs, for transferability more effort is required

- Maintain sparsity of solution
- Regularization via training algorithm



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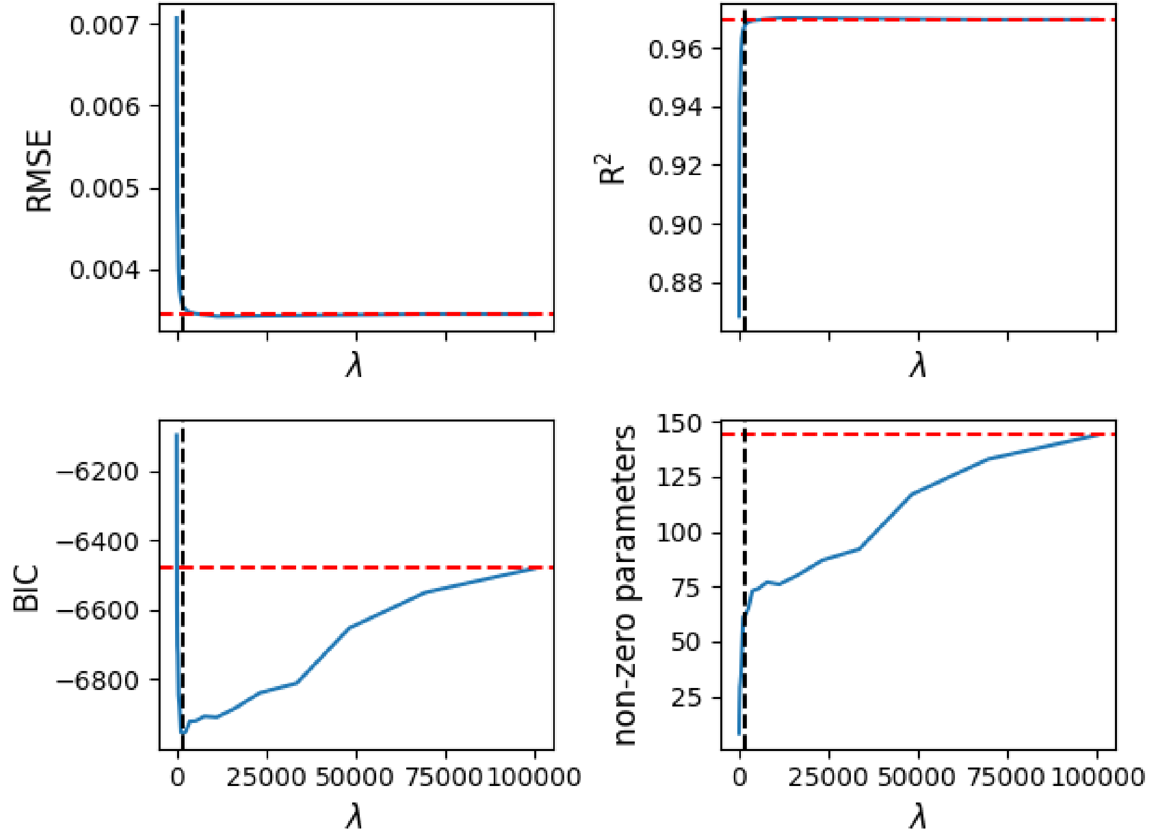
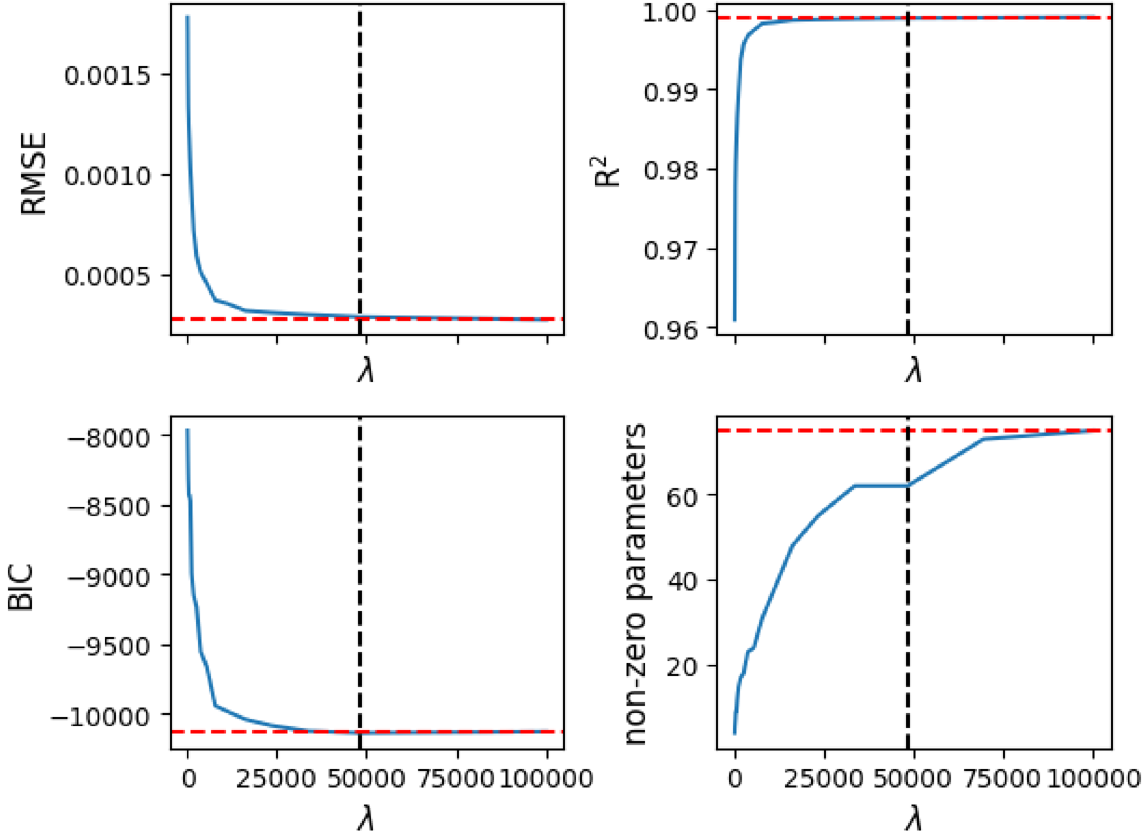
Regularization (ardr)

CuAu



CuNi

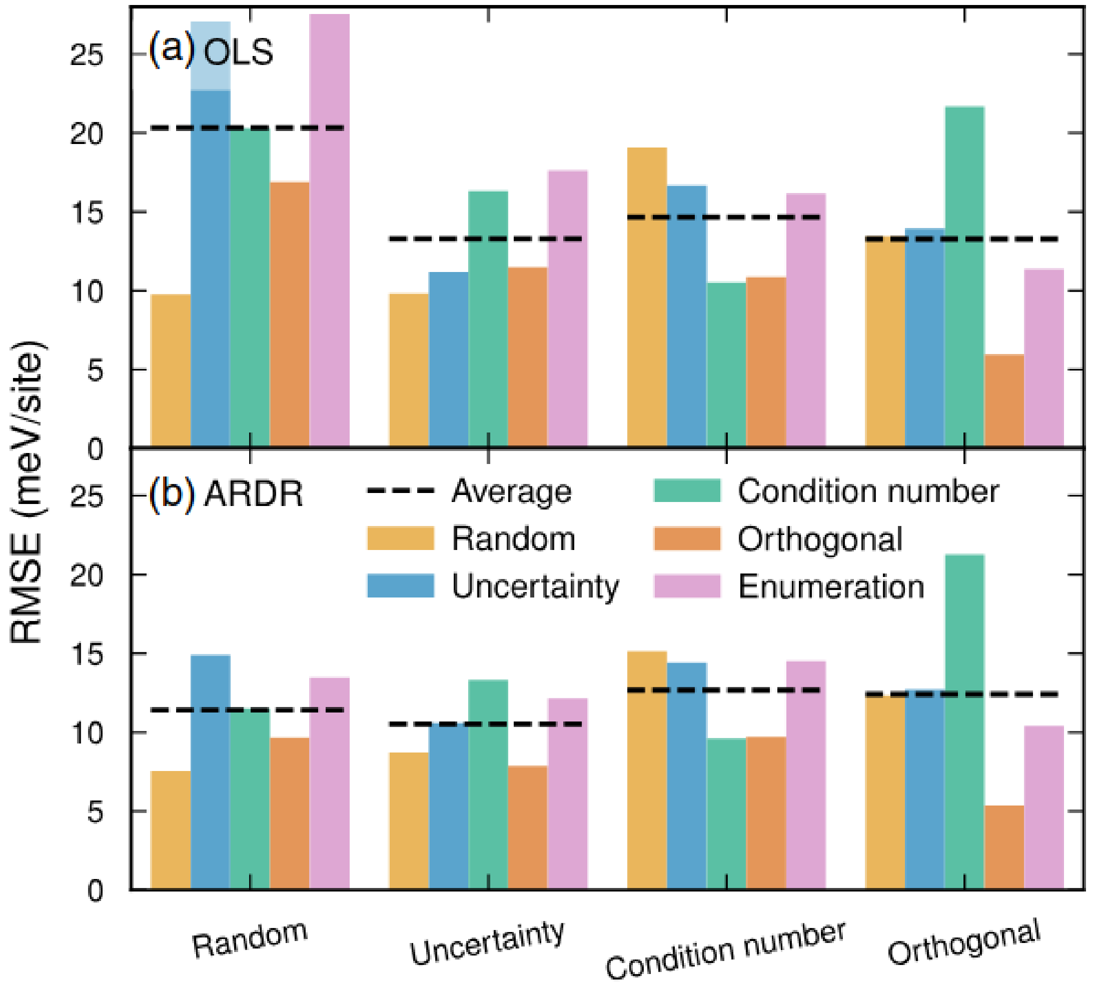


Advanced dataset generation methods

Easy method: enumeration of all structures up to a certain size.

For complex structures:

- Enumeration might not be enough
- More stochastic dataset generation can be required



P. Ekborg-Tanner, P. Rosander, E. Fransson, and P. Erhart, “Construction and Sampling of Alloy Cluster Expansions---A Tutorial,” *PRX Energy*, vol. 3, no. 4, p. 042001, Oct. 2024, doi: 10.1103/PRXEnergy.3.042001.

CE summary

Advantage:

- Extremely fast (especially for compositional changes)

Drawbacks:

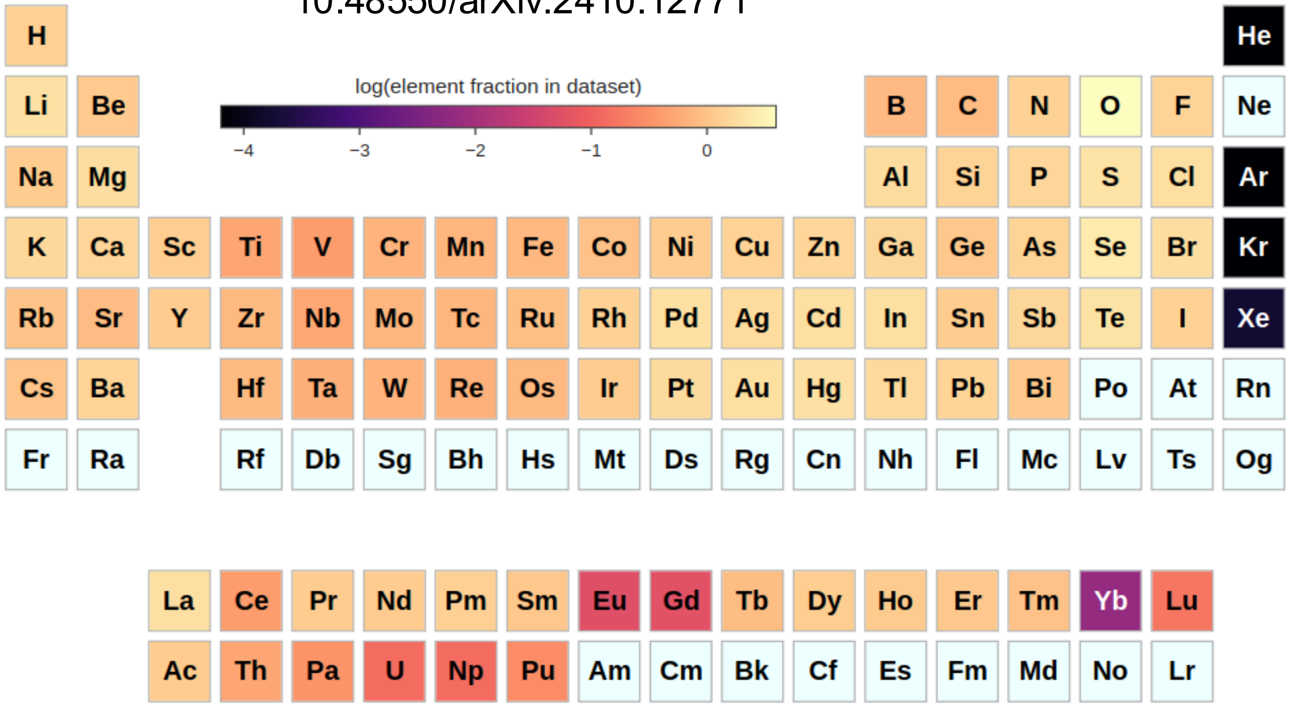
- Can only predict structures relaxing from on-lattice sites
- Only predicts **one** per-atom quantity
- Does not predict actual structure
- Training can be challenging for complex systems

For the workshop

- We use a MACE foundation model to generate the reference data
- Enumerating structures up to a certain size (8 atoms seems to be a good efficiency tradeoff)
- Choose your favorite 2 FCC metals and mix them together
- The CuNi and CuAu datasets are in the repository in case something doesn't work

OMAT24 dataset chart

10.48550/arXiv.2410.12771



See wikipedia for crystal structures